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1. Evaluate each of the following.

   (a) **1/5** $\dfrac{d}{dx}\left[\displaystyle\int_2^x \ln\left(1+t^2\right)dt\right]$

   (b) **2/5** $\displaystyle\int_{-1}^{x} \frac{d}{dt}\left[\frac{t^5}{1+t^6}\right]dt$

   (c) **3/5** $\dfrac{d}{dx}\left[\displaystyle\int_x^3 e^{t^2}dt\right]$

   (d) **4/5** $\dfrac{d}{dx}\left[\displaystyle\int_{-2}^{4x} \cos\left(t^3\right)dt\right]$

2. Evaluate each integral. Hint: remember you can check your answer by differentiating.

   (a) **2/5** $\displaystyle\int \cos\left(8-3x\right)dx$

   (b) **3/5** $\displaystyle\int \frac{dx}{16x^2+1}$

   (c) **4/5** $\displaystyle\int \frac{x^3dx}{2x^4+9}$

   (d) **5/5** $\displaystyle\int_{\frac{2}{\pi}}^{\frac{4}{\pi}} \frac{\sin\left(\frac{1}{x}\right)}{x^2}dx$